

Welcome to BILD62

Dr. Juavinett
jah-vah-nett
(or, Dr. J)



Image: [garetsworkshop/Shutterstock](https://www.shutterstock.com/image-photo/grey-tabby-cat-laptop-keyboard)

Objectives for today

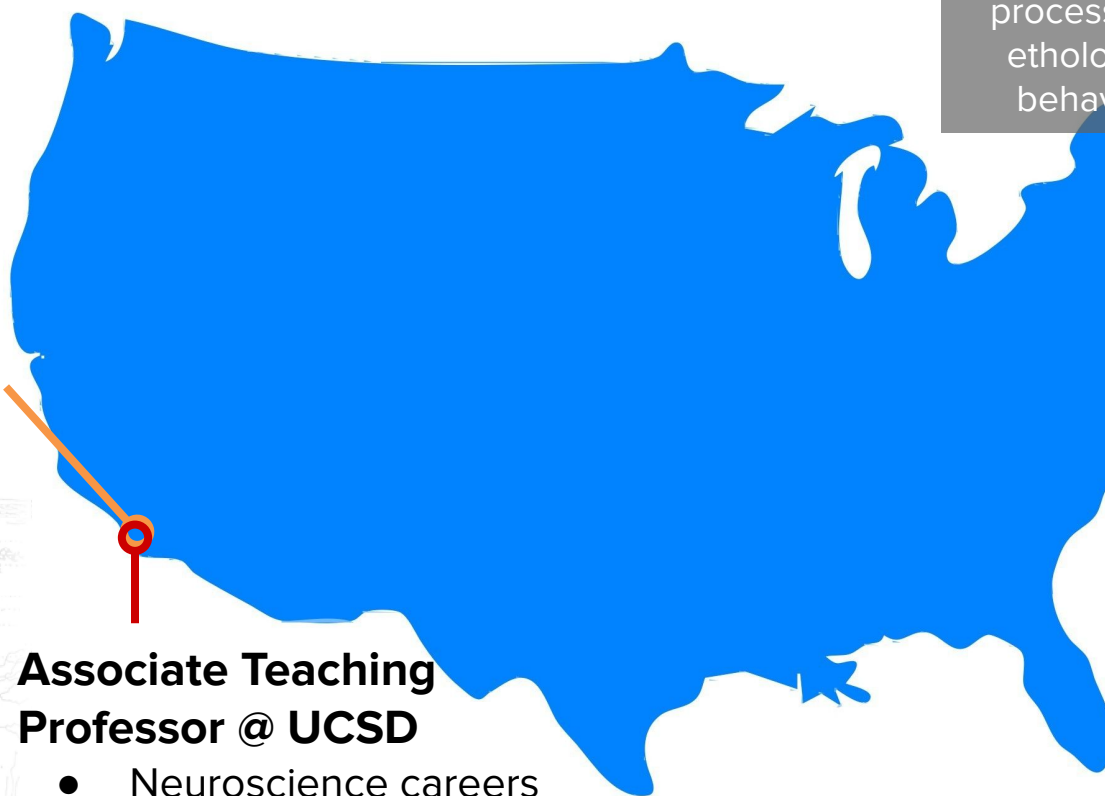
- Introduce the teaching staff, students, and class
- Motivate learning how to code as a biology student
- Discuss course logistics, expectations, & tools
- Start coding!

multisensory
processing &
ethological
behaviors

**Postdoc @
Cold Spring
Harbor
Laboratory**

**Lafayette
College**

**Born in
Philly**

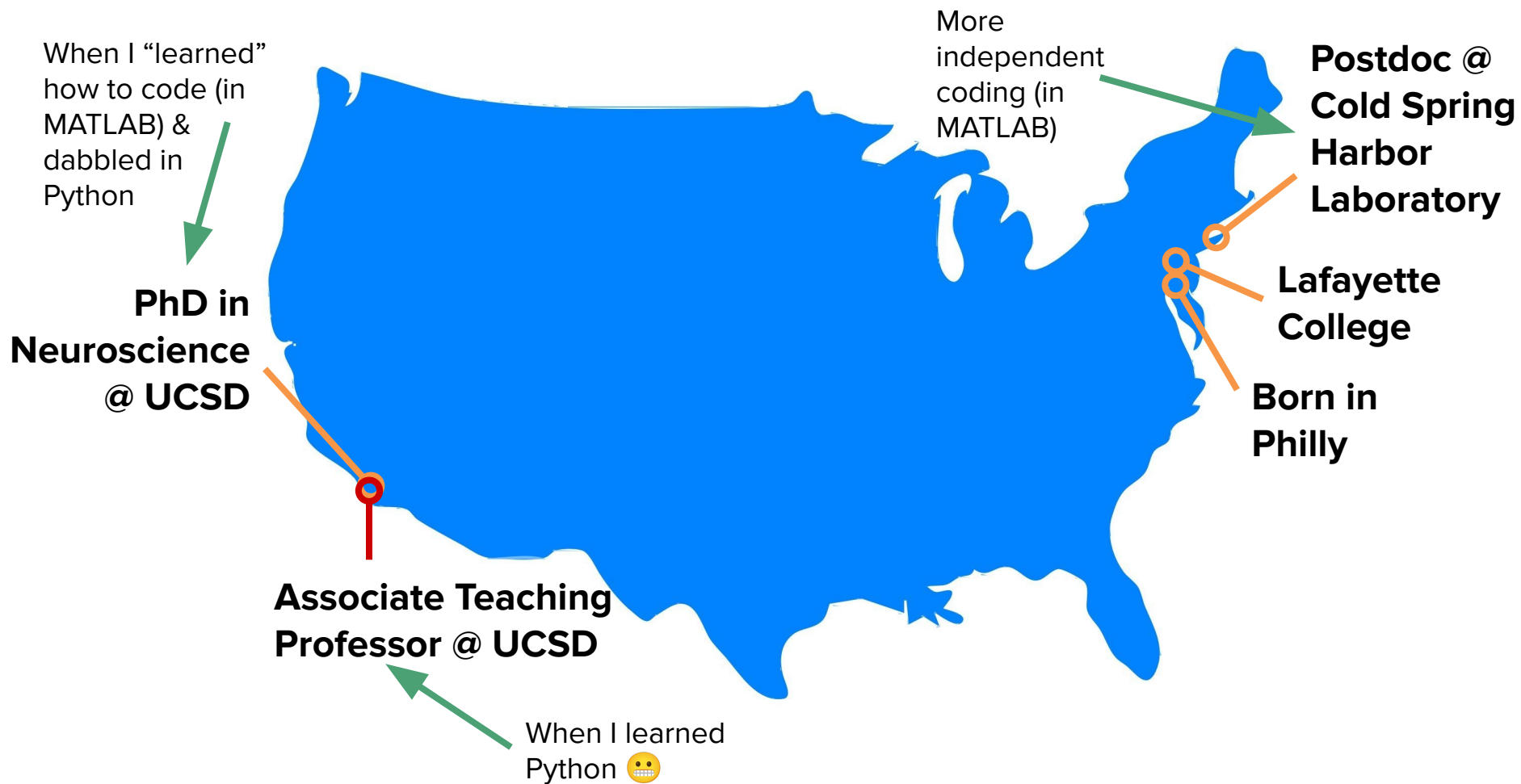


 **PhD in
Neuroscience
@ UCSD**

visual cortex

**Associate Teaching
Professor @ UCSD**

- Neuroscience careers & education
- Open-source data



Introduction to our teaching staff!



TA: Dante Fischer



Reader: Jia Bhavnani



PLA: Izzy Ngyuen

Let's be human, for just a second.

With the folks next to you,
share:

- Your name, major, and preferred pronouns
- Something that brought you joy over the break
- Why you're taking this course





What brought me joy this summer...



Objectives for today

- Introduce the teaching staff, students, and class
- **Motivate learning how to code as a biology student**
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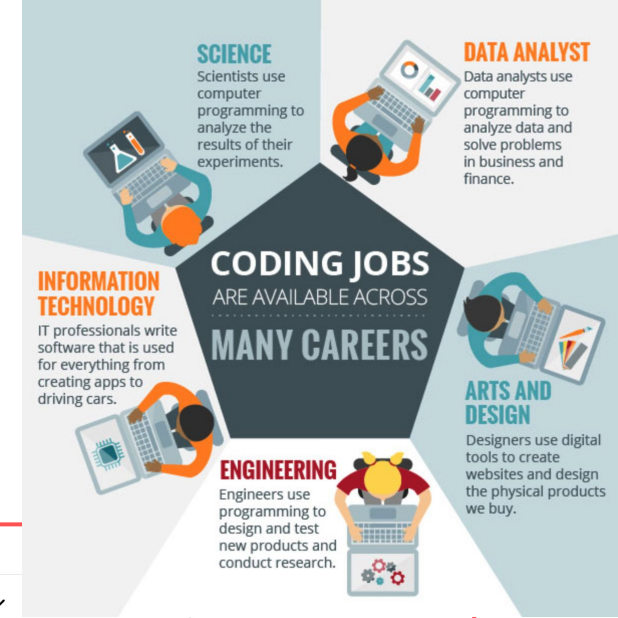
What does coding have to
do with *biology*?

Why *you*, right now?



Why should I learn how to code?

- Programming (and computational skill sets are broadly!) is useful for:
 - Data acquisition (controlling hardware, image acquisition, etc)
 - Data analysis & visualization
 - Computational modeling
- Beyond research, there are jobs for those who have both computational skills *and* disciplinary knowledge →



nature

Explore content ▾ About the journal ▾

[nature](#) > [career q&a](#) > article

CAREER Q&A | 31 May 2022

Why science needs more research software engineers

Ten years after their profession got its name, research software engineers seek to swell their ranks.

By [Chris Woolston](#)

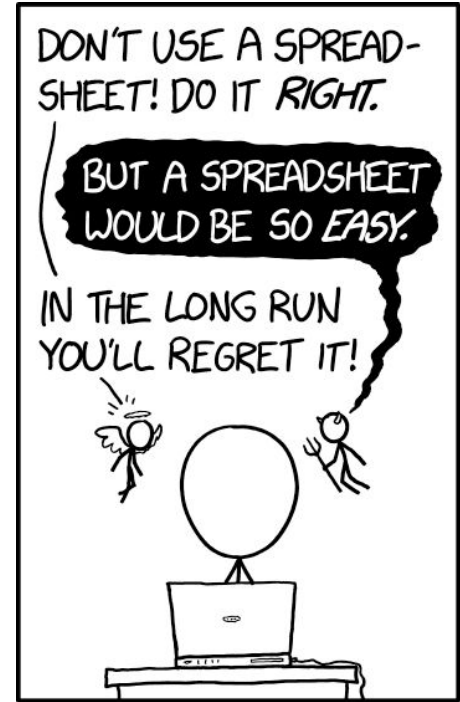
<https://www.nature.com/articles/d41586-022-01516-2>

Excel can only handle datasets with **~1 million rows, and ~16,000 columns** — many datasets in biology are much larger than this!

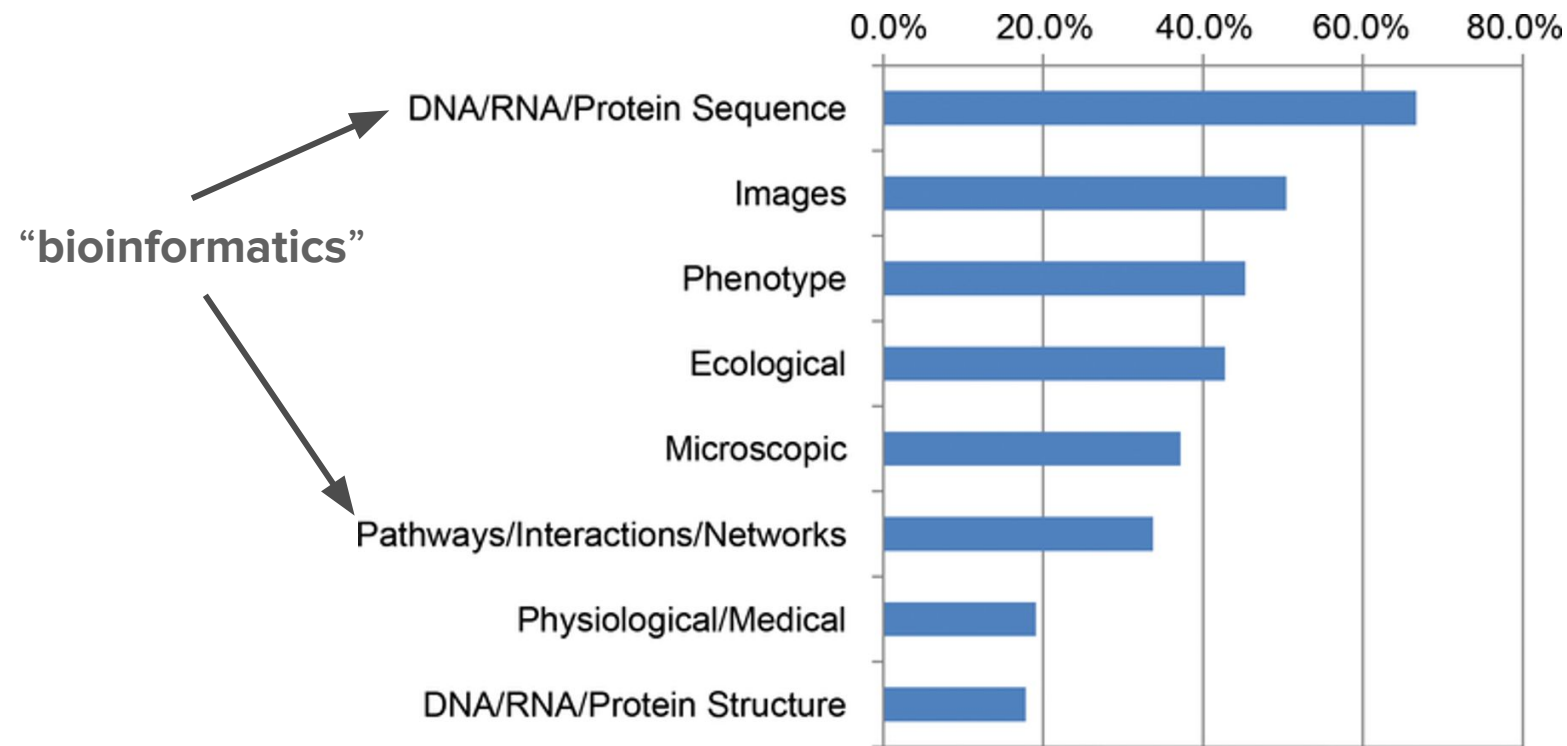
You can automate analyses in Excel, but this is quite limited.

There are also specialized biological data analysis software programs, but often these are limited in how much they can be customized.

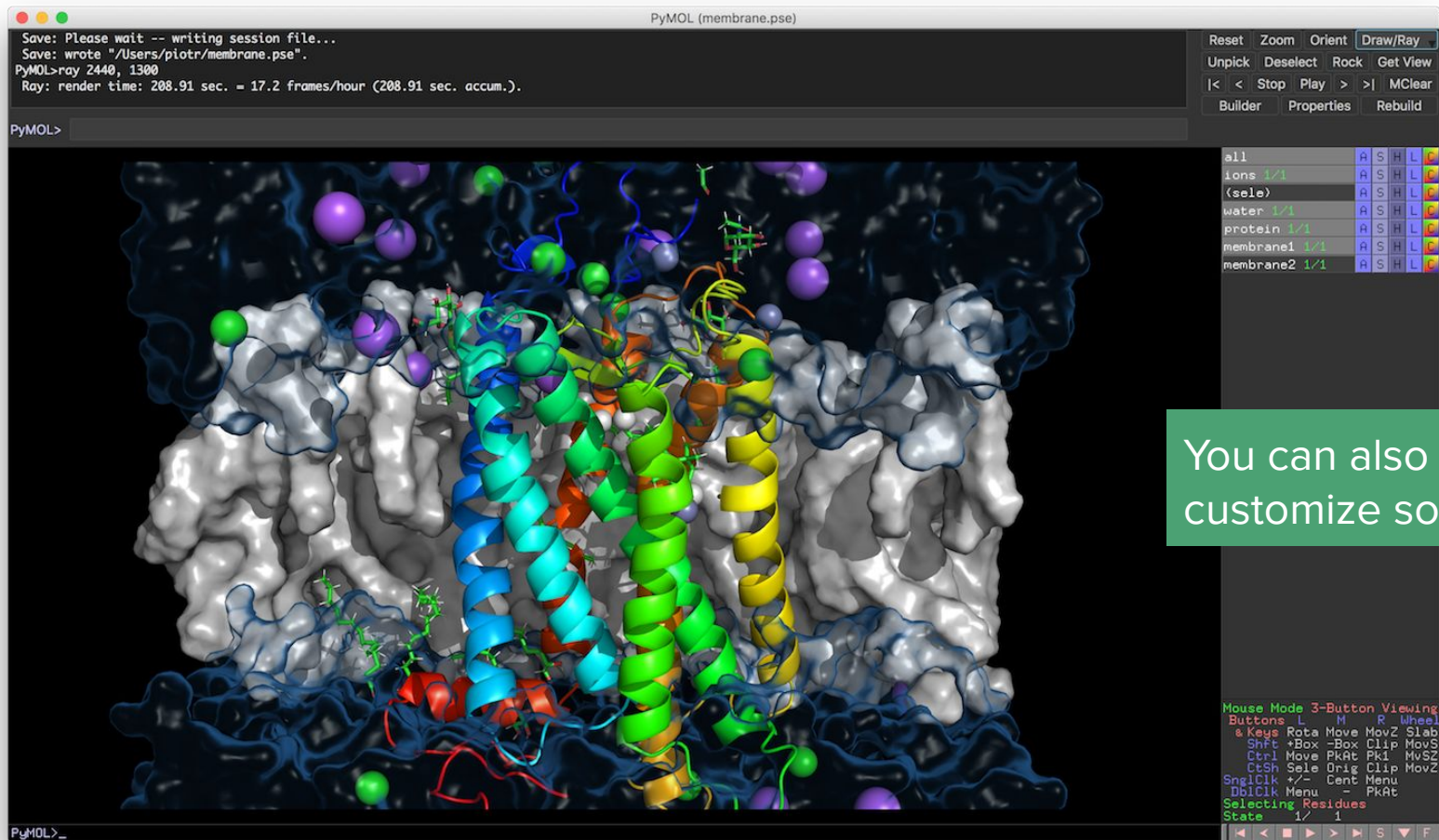
Code is *infinitely* customizable.



<https://xkcd.com/2180/>



Major data types used by National Science Foundation (NSF)
Biological Sciences Directorate principal investigators (PIs).



You can also
customize software!

Open-Source Philosophy

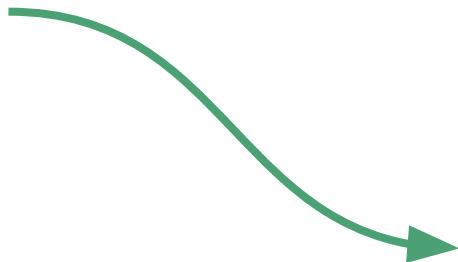
PyMOL is a commercial product, but we make most of its source code freely available under a permissive license. The open source project is maintained by **Schrödinger** and ultimately funded by everyone who purchases a PyMOL license.

Open source enables open science.
This was the vision of the original PyMOL author Warren L. DeLano.

AND many software packages for biologists can be modified... if you know how to code!

Visit the Open-Source Project

Become a sponsor



[Pull requests](#)
[Issues](#)
[Marketplace](#)
[Explore](#)

[schrodinger / pymol-open-source](#)
Public
Watch 32
Fork 166
Star

[Code](#)
[Issues 55](#)
[Pull requests 1](#)
[Discussions](#)
[Actions](#)
[Security](#)
[Insights](#)

[master](#)
3 branches
4 tags
Go to file
Add file
Code

speleo3
iterate: Add explicit_valence and explicit_degree (#227)
 ✓ abc3077 19 days ago
5,076 commits

.github/workflows	CI: Use ubuntu-18.04	16 months ago
contrib	PYMOL-3722 Fix gro file reading	7 months ago
data	PYMOL-3793: Fix for Lighting Plugin on Mac	22 days ago
examples	Fix remaining string module uses	2 years ago
include	pymol::invoke & pymol::apply	last month
layer0	Remove orthoCGO defines; fix warnings	22 days ago
layer1	iterate: Add explicit_valence and explicit_degree (#227)	19 days ago
layer2	iterate: Add explicit_valence and explicit_degree (#227)	19 days ago
layer3	Fix broken group parenting	22 days ago

About

Open-source foundation of the user-sponsored PyMOL molecular visualization system.

[pymol.org/](#)

Readme

View license

634 stars

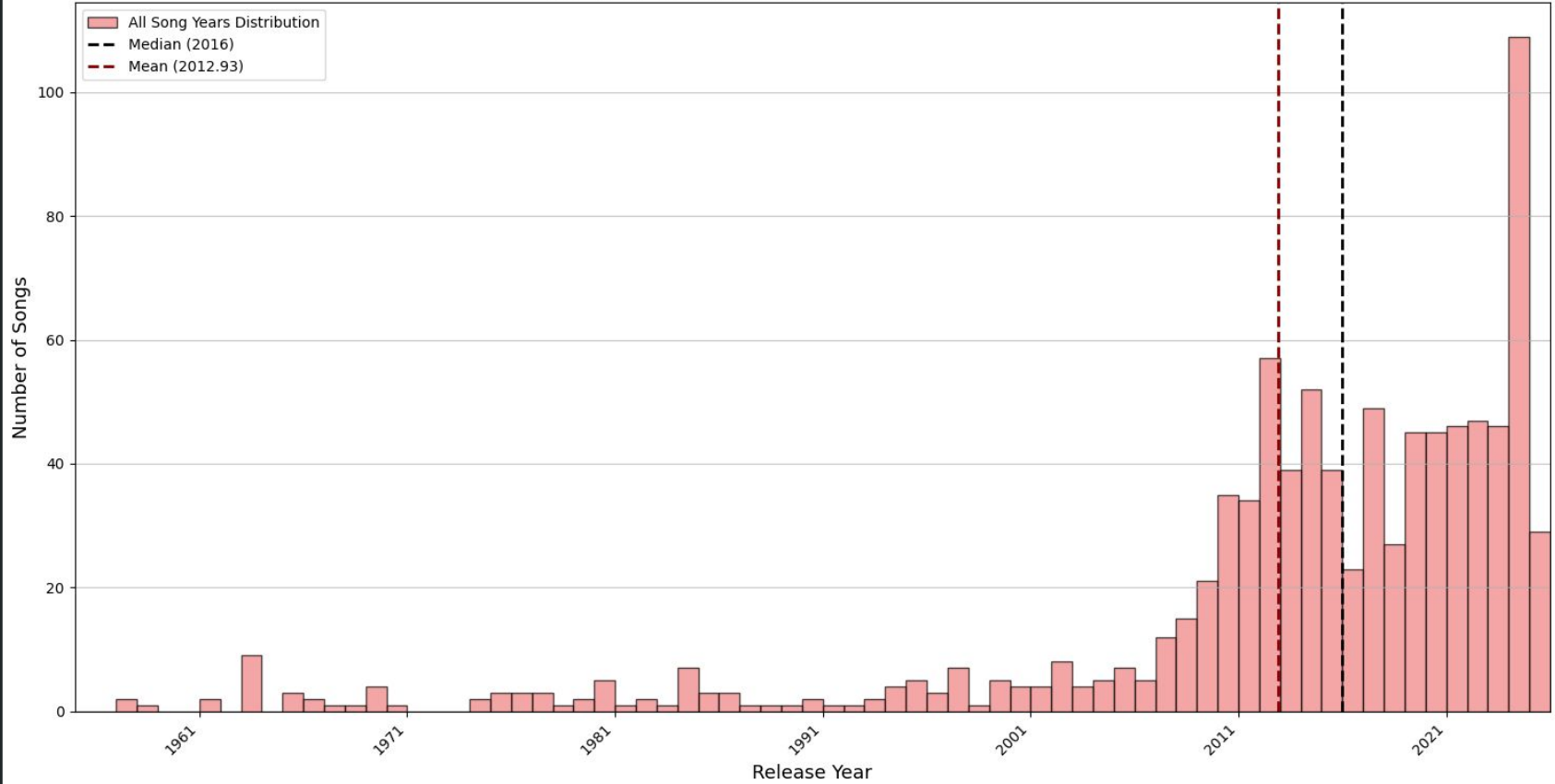
32 watching

166 forks

Releases

4 tags

Histogram of Song Release Years for All Playlists



You can also do things completely unrelated to biology...

 Web API

• Overview

• Getting started

• Building with AI

▶ Concepts

▶ Tutorials

▶ How-Tos

▶ Changelog

 Web API**Retrieve metadata from Spotify content or control playback**

Spotify Web API enables the creation of applications that can interact with Spotify's streaming service, such as retrieving content metadata, creating and managing playlists, or controlling playback.

<https://developer.spotify.com/documentation/web-api>

You can also do things completely unrelated to biology...

Objectives for this morning

- Introduce the teaching staff, students, and class
- Motivate learning how to code as a biology student
- **Discuss course logistics, expectations, & tools**
- Start coding!

First step: let's drop our ideas of what it means to be a ***coder***.

Programming, like learning a language, *takes time*.

Your language brain matters more for learning programming than your math brain

New research contradicts long held assumptions about coding



Amy Nippert

Neuroscience
University of Minnesota

May 12, 2020

 2 peer comments



Christina Morillo on Wikimedia Commons.

<https://massivesci.com/articles/programming-math-language-python-women-in-science/>, summarizes this article: <https://www.nature.com/articles/s41598-020-60661-8>

Previous studies have shown that math and logic problems seem to rely mainly on the multiple demand regions in the left hemisphere, while tasks that involve spatial navigation activate the right hemisphere more than the left. The MIT team found that reading computer code appears to activate both the left and right sides of the multiple demand network, and ScratchJr activated the right side slightly more than the left. This finding goes against the hypothesis that math and coding rely on the same brain mechanisms.

<https://news.mit.edu/2020/brain-reading-computer-code-1215>
about this study: <https://elifesciences.org/articles/58906>

What will help you succeed in this course?

Things that predict success:

- Your belief that you *can* learn how to code
- Completing assignments on time and on your own
- Asking questions when you have them
- Attending discussion sections & office hours
- Testing your knowledge in new situations

Things that ***do not*** predict success:

- Gender
- Age
- Personality
- Math ability
- “Tech-savvyness”



29A @StuxnetStudios · 14h

New programming student:

"I'm not very good at this. When I type out the code, I have to fix lots of errors. And I have to look up how to do most of it."

Instructor:

"You're doing it right."



29



275



1.4K



Historical sidenote: why is it called a **bug**?

In 1947, computer scientist & legend **Grace Hopper** found a *literal bug* in their computer, causing it to produce many errors.



Interview with Grace Hopper:
<https://www.youtube.com/watch?v=QA33wW5LaNY>

Photo # NH 96566-KN (Color) First Computer "Bug", 1947

9/2
9/9

0800 Action started
1000 " stopped - action ✓ { 1.2700 9.037 847 025
1300 (032) MP - MC 1.582 1.000 9.037 846 995 convd
2.130476415 (2) 4.615925059 (-2)
(033) PRO 2 2.130476415
convd 2.130676415
Relays 6-2 in 033 failed special speed test
in relay " 11.00 test.
Relays changed
1100 Started Cosine Tape (Sine check)
1525 Started Multi Adder Test.
1545  Relay #70 Panel F
(moth) in relay.
1630 First actual case of bug being found.
1700 Action started.
1700 closed down.

Relay 3145
Relay 3370

<https://www.nationalgeographic.org/thisday/sep9/worlds-first-computer-bug/>



Historical sidenote:
Women were the first
computers!

Doris Baron, a human
“computer” works with tape from
the air pressure measuring
machines in 1955

The Gendered History of Human
Computers

What is programming, anyway?

- Programming is the way humans communicate with **computers**
 - It's a language!



What is programming, anyway?

- Programming is the way humans communicate with computers
 - It's a language!
- The instructions we give the computer are taken **literally** and

sequentially

Capitalization matters:
`print()` $\neq$ `Print()`

```
b = a * 2
a = 2
```

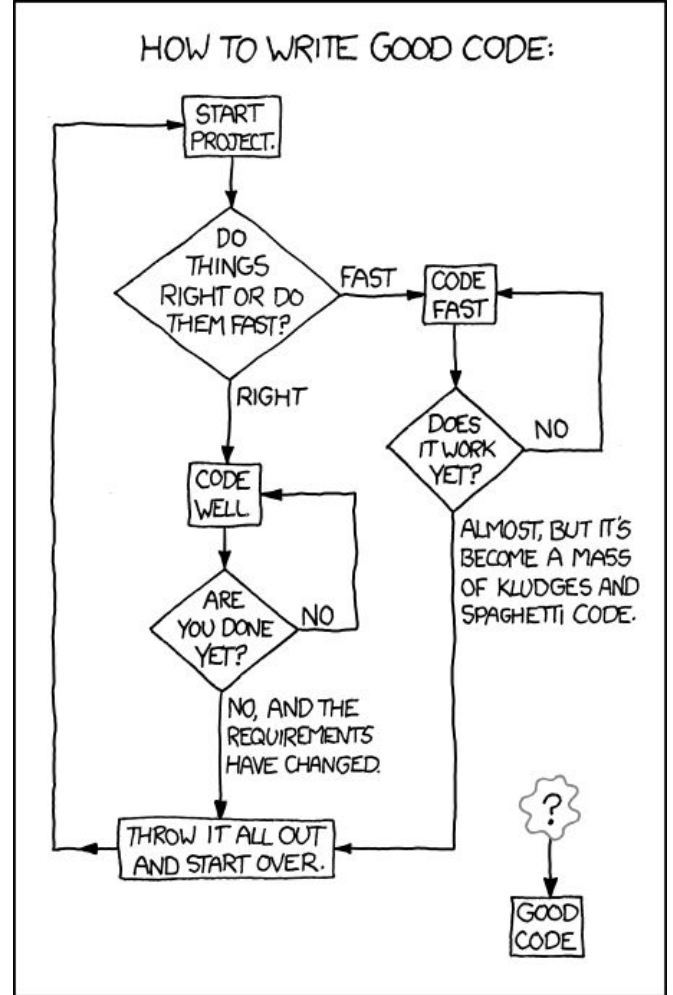
computer: what is a?



What is a “program”?

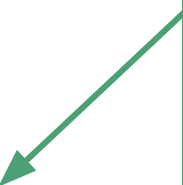
The path to writing good, efficient code

1. Make it **work**
2. Make it **right**
3. Make it **fast**



The path to writing good, efficient code

1. Make it **work**
2. Make it **right**
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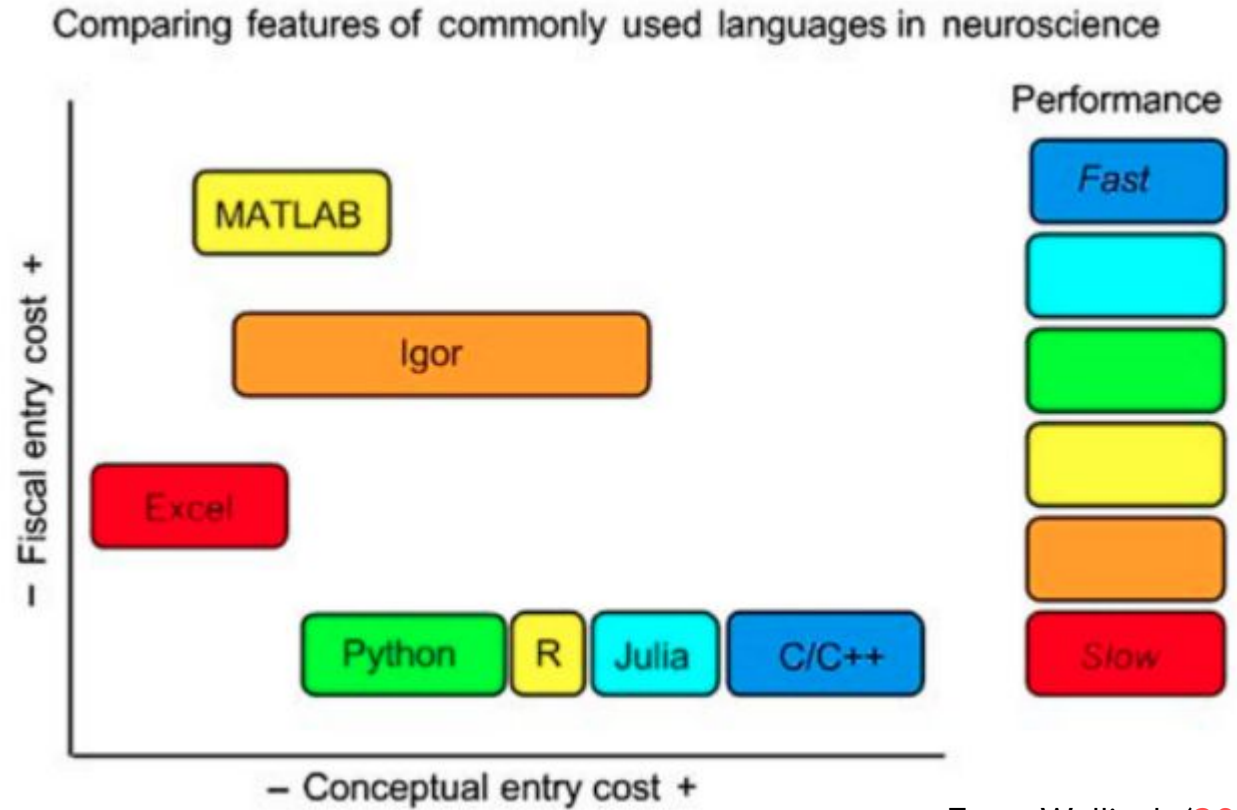
Our goal is to get to this step

If you ultimately became a ***full-time, professional developer***, you'd care about step 3.

For most problems biologists face, step 3 isn't paramount.

Considerations for choosing a programming language

- Fiscal & conceptual entry
- Usage in particular field or profession



From Wallisch ([2017](#))

All coding languages eventually need to talk to the computer in binary:

01001000 01100101 01101100 01101100 01101111 00100001

(hello)

[Learn How To Write Your Name In Binary Code](#)

There are many types of binary code, beyond computers



Braille

<https://www.afb.org/blindness-and-low-vision/braille/what-braille>

A ● -	J ● - - -	S ● ● ●
B - ● ● ●	K - ● -	T -
C - ● - ●	L ● - ● ●	U ● ● -
D - ● ●	M - -	V ● ● ● -
E ●	N - ●	W ● - -
F ● ● - ●	O - - -	X - ● ● -
G - - ●	P ● - - ●	Y - ● - -
H ● ● ● ●	Q - - ● -	Z - - ● ●
I ● ●	R ● - ●	

Morse code

https://www.discoveryworld.org/about/blog/discover_at_home/morse-code/

In this class, we'll use Python

- Programming language, development led by Python Software Foundation (www.python.org)
- Uses concise structure & wording similar to human language
- A “high-level language”



Assembly language vs. high-level language

```
section .text
global _start
_start:
    mov ecx, 10
    mov eax, '0'
l1:
    mov [num], eax
    mov eax, 4
    mov ebx, 1
    push ecx
    mov ecx, num
    mov edx, 1
    int 0x80
    mov eax, [num]
    inc eax
    pop ecx
    loop l1
    mov eax, 1
    int 0x80
section .bss
    num resb 1
```

```
for num in range(10):

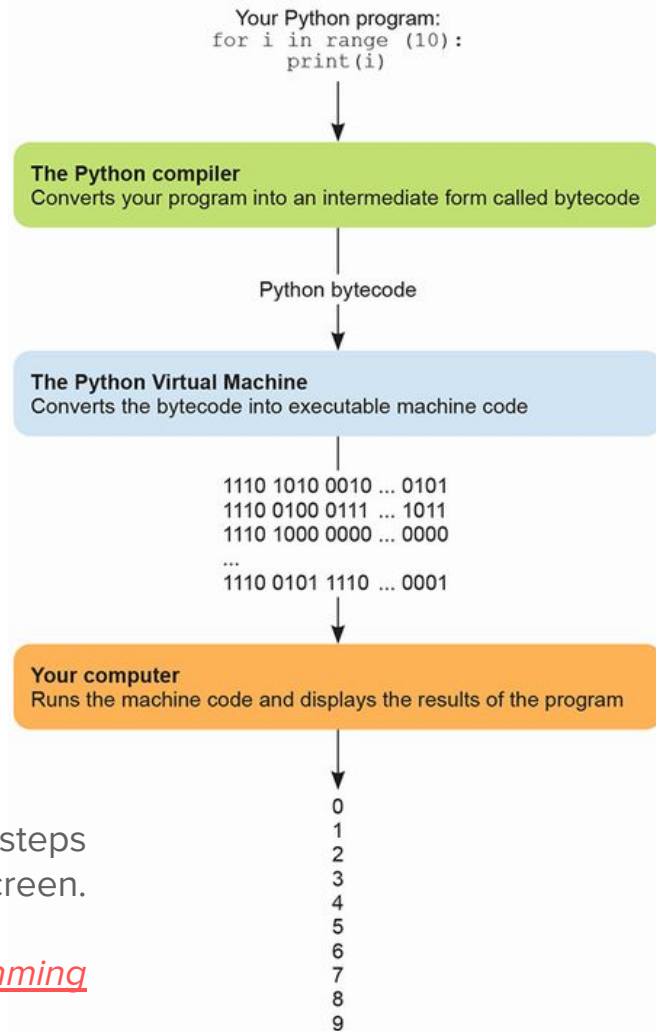
    print(num)
```

In this class, we'll use Python

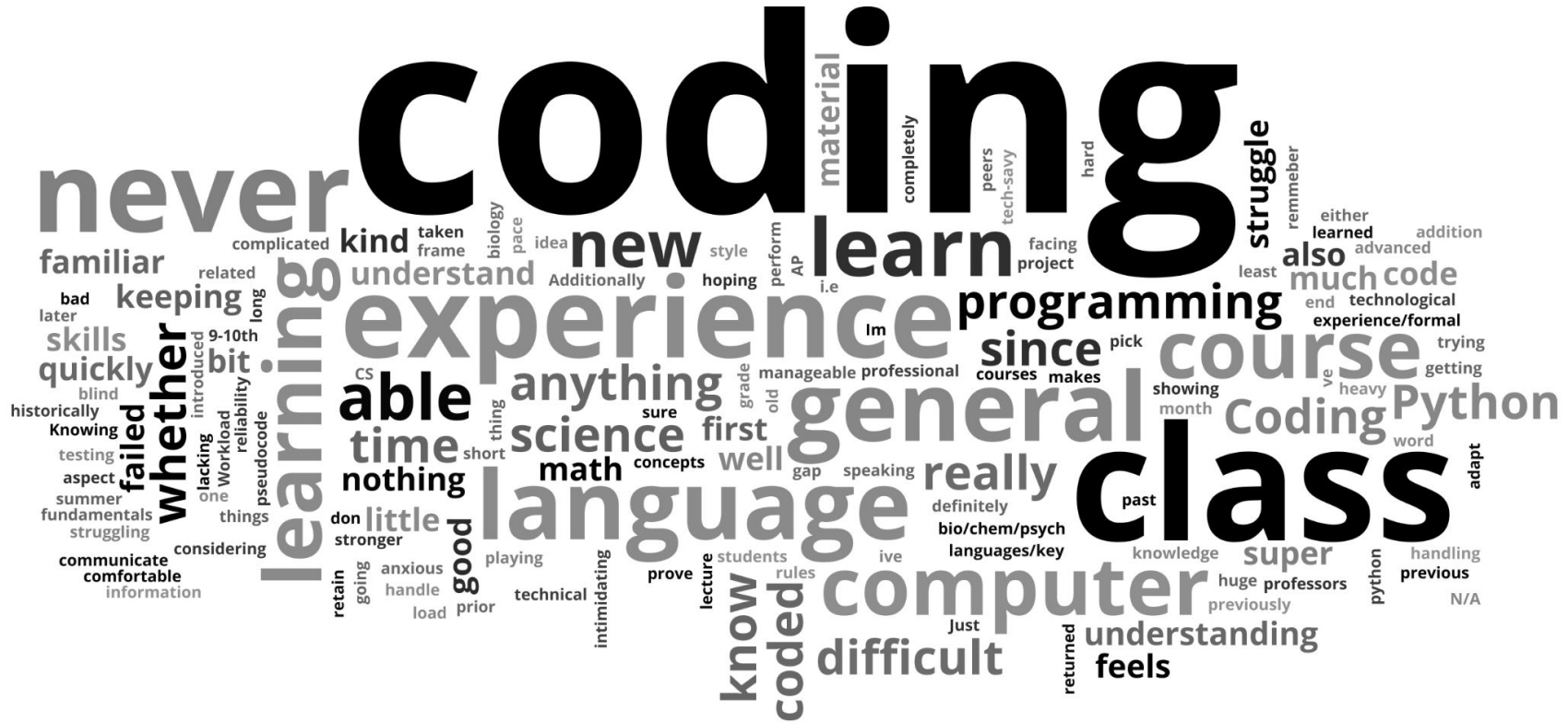
- Python can be used for many purposes, from web programming, to creating games, to analyzing & visualizing data
 - Extension: '.py'
- We'll primarily work in **Jupyter Notebooks**
 - Extension '.ipynb'

Your Python program goes through several steps before you see the output on your screen.

From Porter & Zingaro, [Learn AI-Assisted Python Programming](#)



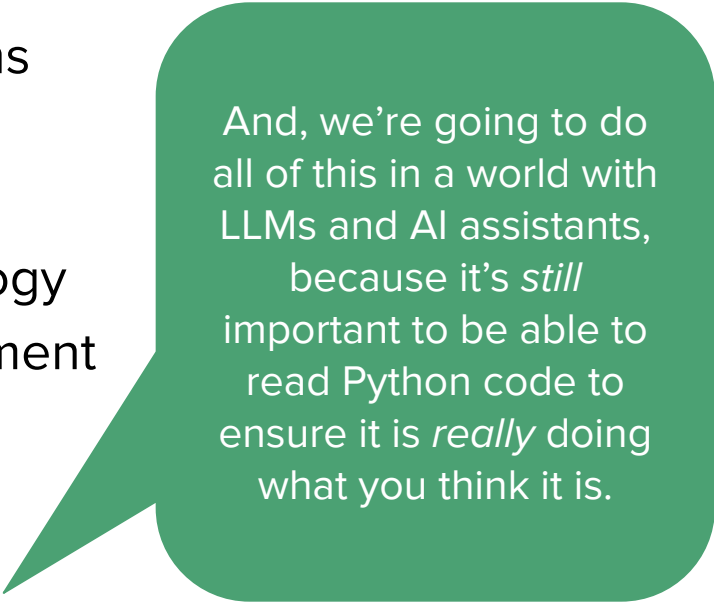
What are you worried about?



A word cloud visualization representing concepts associated with learning biology through coding. The largest and most central words are "coding", "understanding", "python", "biology", "science", "knowledge", "research", "analysis", "programing", and "experience". Other significant words include "use", "basic", "fundamentals", "skills", "want", "like", "important", "apply", "also", "deeper", "fully", "field", "know", "tools", "make", "able", "intersection", "least", "biological", "regarding", "main", "help", "single", "way", "Principles", "foundational", "aspects", "solve", "useful", "liking", "genes", "related", "projects", "logic", "actually", "learns", "teach", "vague", "changed", "broader", "applied", "collection", "feel", "looking", "Learn", "comfortable", "something", "idea", "Application", "aspect", "analyzing", "applications", "Pseudocode", "taken", "intersections", "fairly", "chaotic", "improvement", "suck", "especially", "Java", "much", "molecules", "language", "insight", "good", "world", "interested", "CS", "trying", "hard", "fields", "speed", "gather", "ability", "little", "algorithms", "even", "write", "Learning", "I.e.", "compile", "Python", "lab", "increasingly", "languages", "compute", "Insight", "General", "computers", "expression", "everyday", "efficiently", "generating", "Coding", "computer", "working", "examples", "quicker", "hands-on".

Course Objectives

- Explain, edit, and execute Python programs
- Recognize, create, and manipulate data structures and other code constructs
- Identify use cases of programming in biology
- Collaboratively design programs to implement biology-related workflows
- Develop metacognition around using AI assistance for programming



And, we're going to do all of this in a world with LLMs and AI assistants, because it's *still* important to be able to read Python code to ensure it is *really* doing what you think it is.

Grading breakdown

- **Exams (50%)**
 - Three in person quizzes (10% each) that will test you on your understanding of class material.
 - Cumulative in-person final (20%) at the end of the quarter.
- **Final Projects (25%)** Includes the project proposal, code, and deliverables.
- **Assignments (25%)**

Assignments & project components **lose 10% each day they are late.**

You can arrange to take any of these exams at the Triton Testing Center if needed (and are encouraged to do so early in the quarter).



Assignments

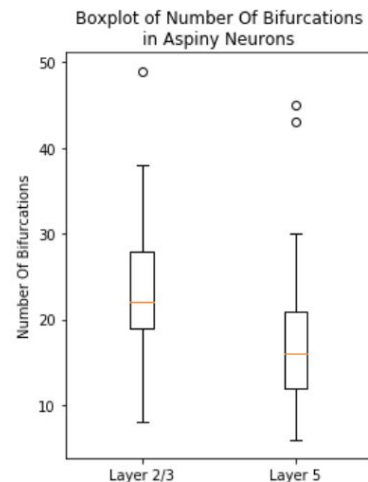
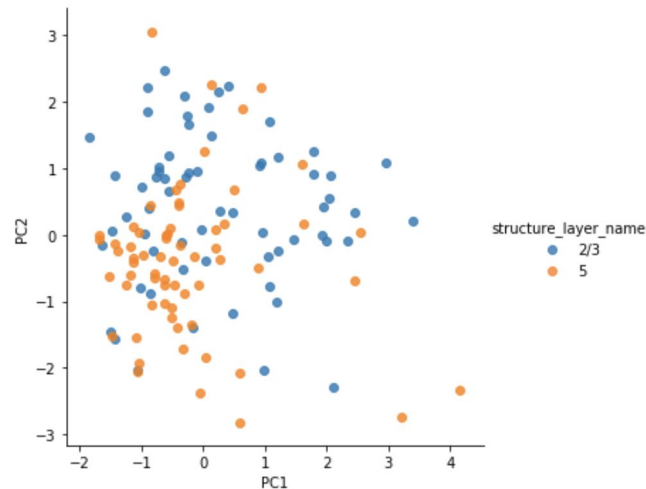
In-Class Work (15%): Occasionally in class, you will complete activities for credit. You need to be in class to participate. Your lowest two assignments in this category will be dropped.

Discussion Attendance (5%): You will receive credit for completing weekly questions during your discussion section. If you complete 6/8 of these, you'll receive full credit for this portion of your grade.

Practice Quizzes (5%): These quizzes will be posted before each of our quizzes and the final.

Final Project

- Completed in groups of 3
 - Your group will be automatically assigned based on the discussion section *that you regularly attend*.
- Includes the project proposal & final deliverable (code and descriptive text)
- Your final project will either:
 - Write a program to complete a task
 - Analyze & visualize data of your choosing
- We will discuss possibilities for your project as we move through the course.



END OF YEAR SALE - SAVE 50%

0 1
Days

0 6
Hours

1 1
Minutes

0 6
Seconds

VIEW PLANS



DATAQUEST

COURSES

STUDENT STORIES

WE'RE HIRING

BLOG

START LEARNING

LOG IN

Learn Data Science

Whether you're new to the field or looking to take a step up in your career, Dataquest can teach you the data skills you'll

Take a FREE course!

Email

Password

There are also lessons via an online **Intro. to Comp. Sci. & OOP: Python** course (<https://cogniterra.org/course/326/promo>) or **DataQuest** (free!) & complete lessons in parallel with our course.

Python Basics for Data Analysis (Skill Path) or Data Scientist in Python (Career Path)

Office hours

Both Dr J and our PLA will have office hours.

Vote on the entry survey!

Why should you come to office hours?

- You have clarifying questions about the course or its content
- You have concerns about the course and your progress
- You'd like to talk about career paths in biology or neuroscience



Course Tools



canvas

Submitting non-coding
assignments & managing grades



“Tutoring” and project help;
more on this later.

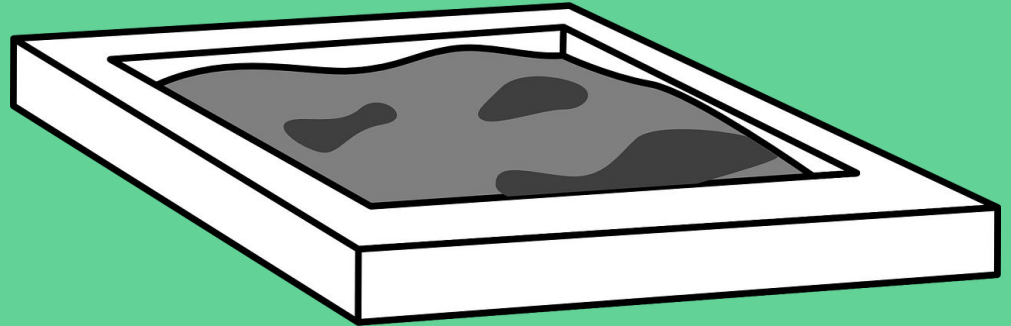


Coding exercises & assignments



Sharing public course materials
<https://github.com/BILD62>

Interacting with course materials



You can find all of our course materials on either Canvas or the course GitHub

Lectures

In other words, PDF slides shown during class.

Hosted on GitHub in the Lectures folder

If I use both a PDF and a Jupyter Notebook during lecture, these numbers will match

Materials

Jupyter Notebooks

You can pull these locally or to DataHub, or look at them online via GitHub or Colab/ Binder



Assignments

Jupyter Notebooks, submitted through **Assignments** tab

Answers posted in the Assignments folder on Github



THE MAGIC LINK FOR THIS COURSE:

Sync with your datahub:



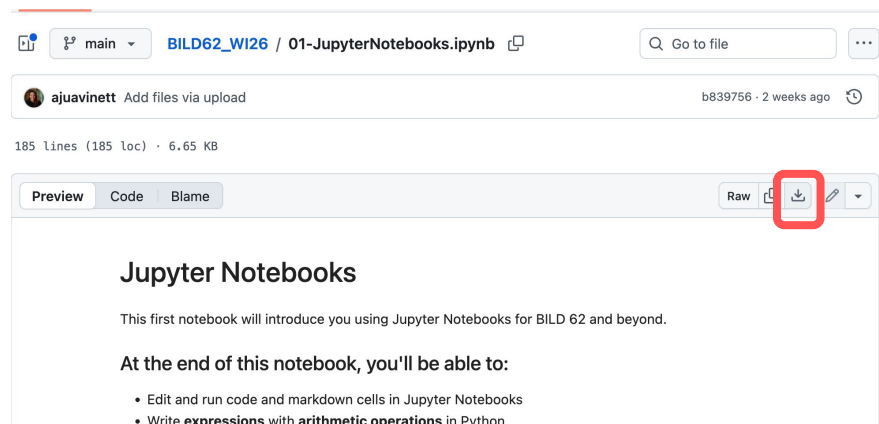
[https://datahub.ucsd.edu/hub/user-redirect/
git-sync?repo=https://github.com/BILD62/BILD62_FA26](https://datahub.ucsd.edu/hub/user-redirect/git-sync?repo=https://github.com/BILD62/BILD62_FA26)



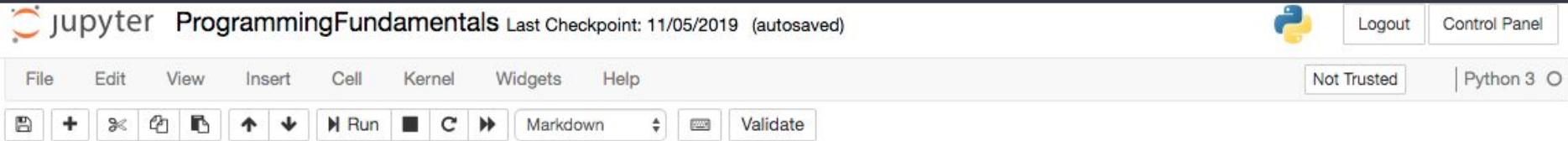
*Where our course
content lives*

To clone Materials to DataHub:

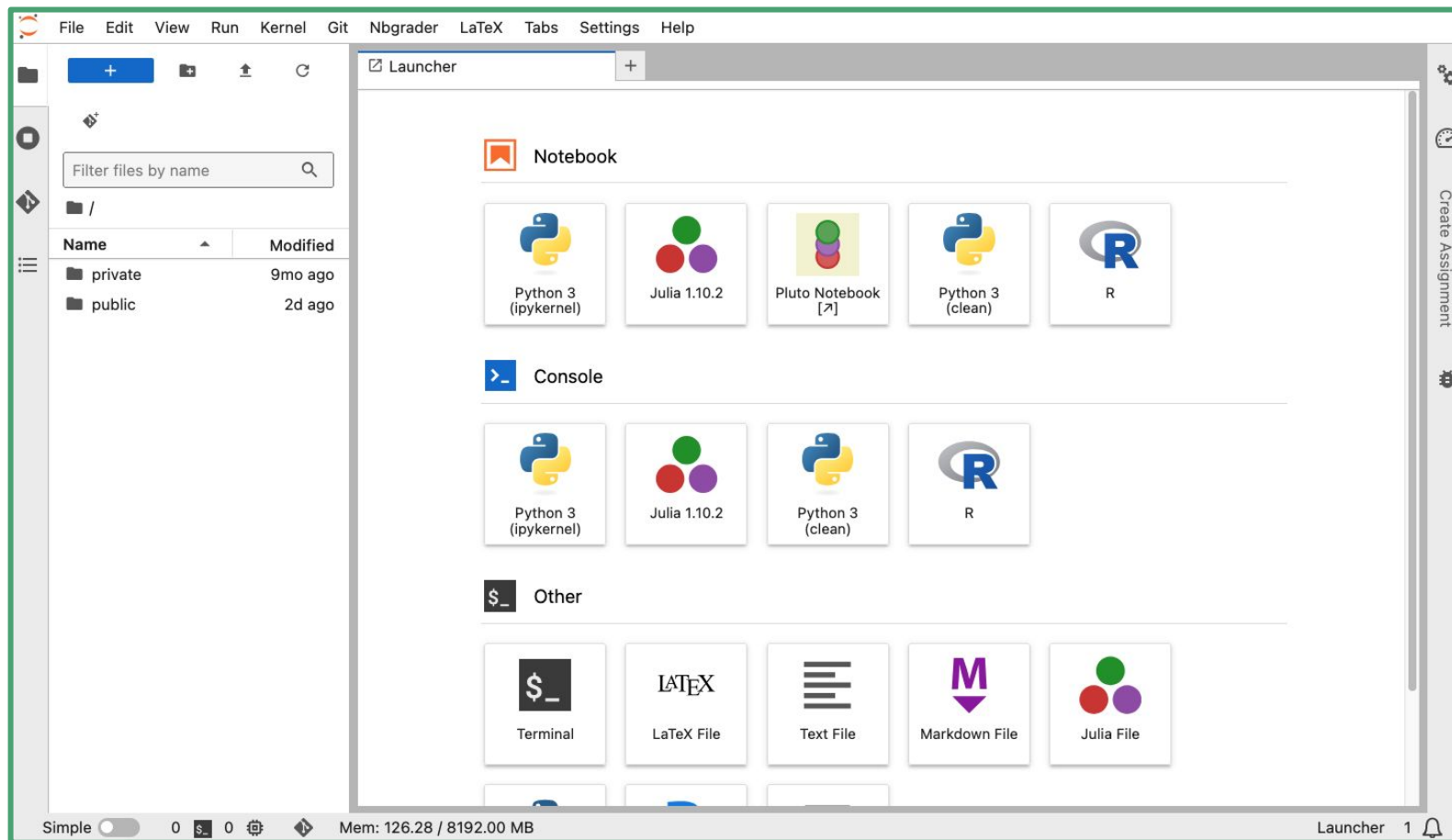
1. Click on the magic link.
2. Log in to DataHub as prompted.
3. You'll see the newest version of the course materials in your DataHub!
 - **If you don't see the course materials, click the magic link one more time.**
4. If you want a fresh copy of the materials, you can manually download it from our Github & upload to DataHub.



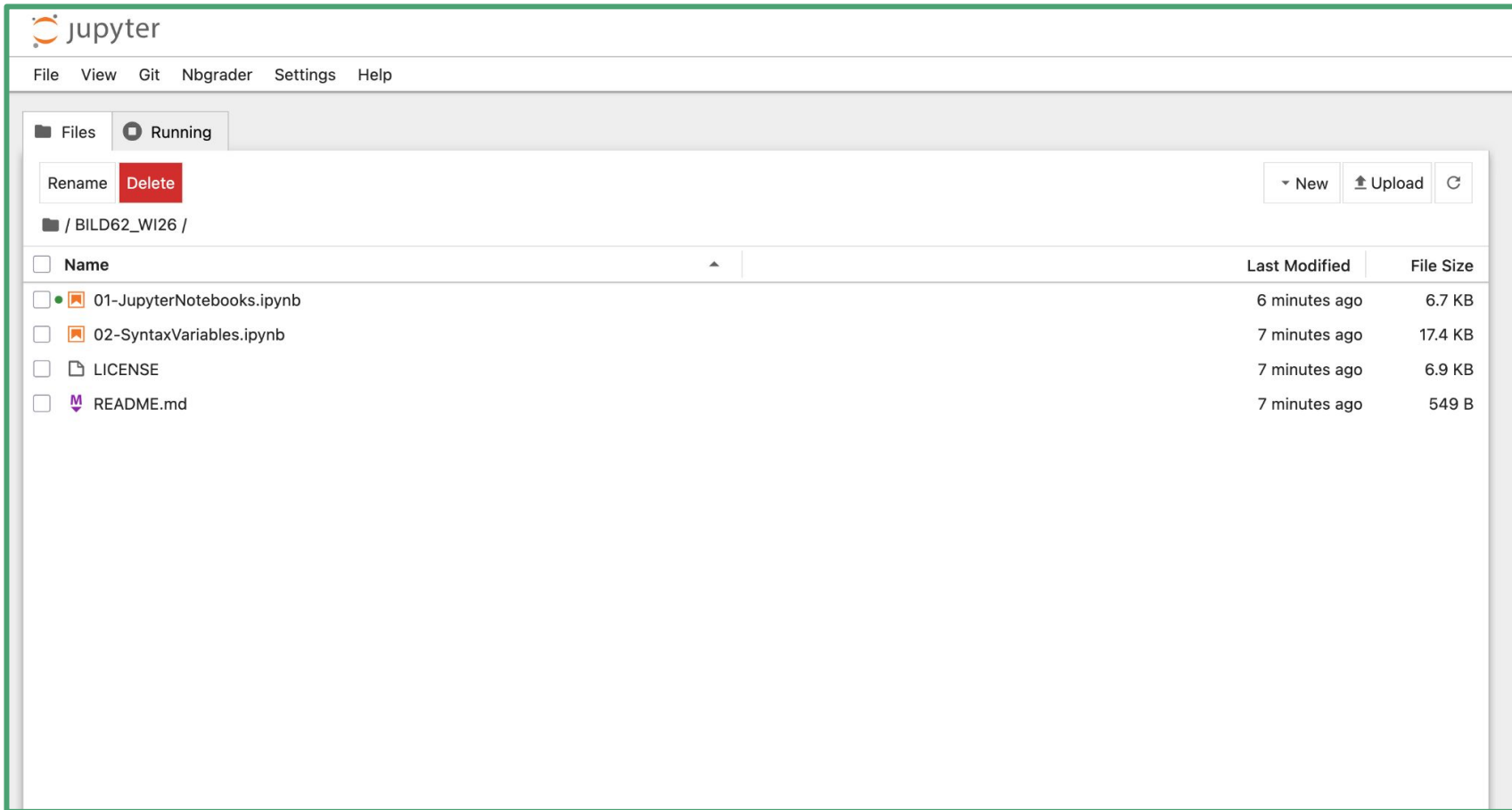
Introduction to the UCSD DataHub & Jupyter Notebooks



The **launcher**: what you'll see if you launch from datahub.ucsd.edu



file browser view: what you'll see if you launch from the magic link



The screenshot displays the Jupyter file browser interface. At the top, the Jupyter logo and name are visible. Below the logo is a navigation bar with links for File, View, Git, Nbgrader, Settings, and Help. The main area shows a directory listing for /BILD62_WI26/. The interface includes tabs for Files and Running, with the Running tab currently selected. Action buttons for Rename and Delete are present, along with buttons for New, Upload, and Refresh. The file listing table shows four items: 01-JupyterNotebooks.ipynb (6.7 KB, 6 minutes ago), 02-SyntaxVariables.ipynb (17.4 KB, 7 minutes ago), LICENSE (6.9 KB, 7 minutes ago), and README.md (549 B, 7 minutes ago).

jupyter

File View Git Nbgrader Settings Help

Files Running

Rename Delete

New Upload Refresh

/ BILD62_WI26 /

☐ Name	Last Modified	File Size
☐ 01-JupyterNotebooks.ipynb	6 minutes ago	6.7 KB
☐ 02-SyntaxVariables.ipynb	7 minutes ago	17.4 KB
☐ LICENSE	7 minutes ago	6.9 KB
☐ README.md	7 minutes ago	549 B

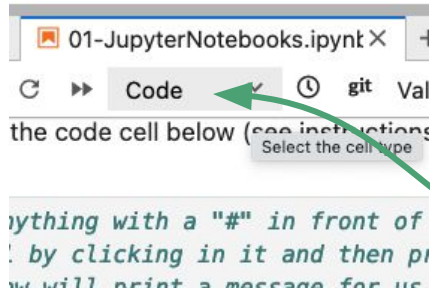
About Jupyter Notebooks

- Jupyter is a loose acronym for Julia, Python, and R
- Run in a web browser but it's not *necessarily* online (it *is* online when we use the DataHub)
- Usefully, it will show plots directly in the notebook as you work your way through, performing analyses in real-time (this is why it is used by many scientists!)
- **If you change anything in the cell, you need to re-run it.**



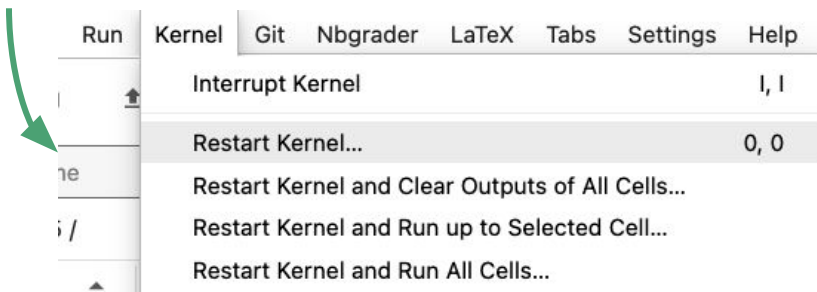
Using Jupyter Notebooks

- **Cell**: the main organizational structure of the notebook
 - Use **Shift+Enter** to run a cell (or press Run)
 - You can run cells out of order, and move cells around!
 - Cells can be **code** (the default) or **markdown** (descriptive text or images)
 - Code cells have `[ ]`:
 - If there is a star (`[ ] * :`), that means your cell is running
 - Change between code & markdown using **Code** menu (or keyboard shortcuts)



Kernel: the engine that runs the code

- You can clear your namespace and get a fresh start by restarting the kernel
- Use **Kernel** menu to interrupt and/or restart if it gets stuck!



You can tell if the kernel is busy by whether or not the circle next to Python 3 (upper right corner) is filled or not. (filled = busy)

Expressions describe
how to combine pieces of
data (e.g., add them!)

SYMBOLS YOU WILL ENCOUNTER IN THIS COURSE

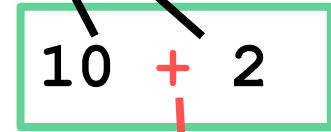
Symbol	Name	Sample Usage
=	Equal sign	Assign variable
#	Pound sign; hashtag	Line comments
[]	Brackets	Indexing & Slicing
()	Parentheses	Using functions
{ }	Curly Brackets	Defining dictionary
' '	Single quotes	Creating string
" "	Double quotes	Creating string
_	Underscore	In variable names
!	Explanation point	To test not equal (!=)
\	Back slash	Delineate line break
:	Colon	Indexing

Basic arithmetic operators in Python

Symbol	Operation	Usage
+	Addition	10+2
-	Subtraction	10-2
*	Multiplication	10*2
/	Division	10/2
**	Exponent	10**2
%	Modulo	10%2

inputs

expression



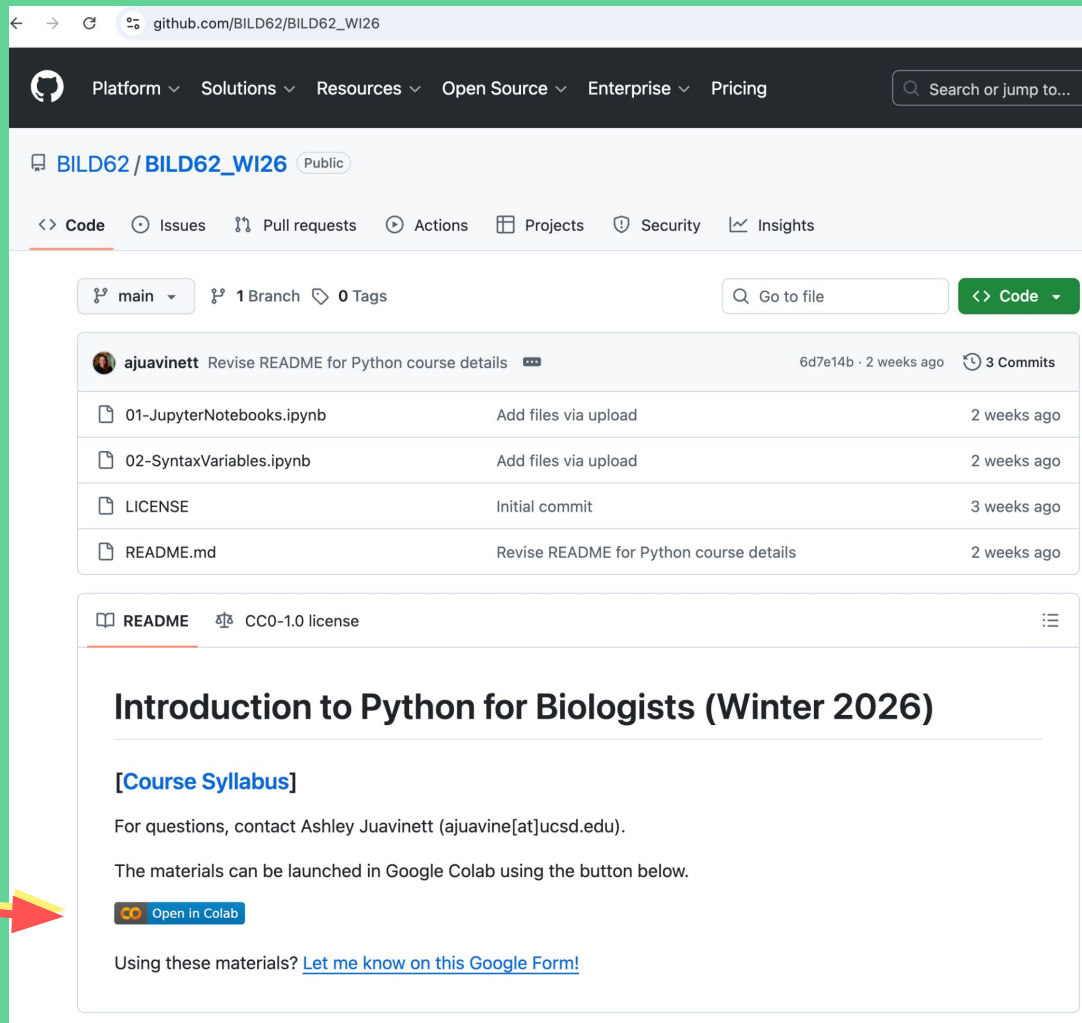
operand

If you want a whole number (floor division), use // instead.

Let's code!

Use the magic link (on Canvas and these slides) to sync up your DataHub with our folder, and open notebook 01.

If DataHub isn't working, use the "Open in Colab" link on our GitHub



github.com/BILD62/BILD62_WI26

Platform Solutions Resources Open Source Enterprise Pricing

BILD62 / BILD62_WI26 Public

Code Issues Pull requests Actions Projects Security Insights

main 1 Branch 0 Tags

Go to file Code

ajuavinett Revise README for Python course details 6d7e14b · 2 weeks ago 3 Commits

01-JupyterNotebooks.ipynb	Add files via upload	2 weeks ago
02-SyntaxVariables.ipynb	Add files via upload	2 weeks ago
LICENSE	Initial commit	3 weeks ago
README.md	Revise README for Python course details	2 weeks ago

README CC0-1.0 license

Introduction to Python for Biologists (Winter 2026)

[\[Course Syllabus\]](#)

For questions, contact Ashley Juavinett (ajuavine[at]ucsd.edu).

The materials can be launched in Google Colab using the button below.

[Open in Colab](#)

Using these materials? [Let me know on this Google Form!](#)

Before next class...

- Access Canvas (canvas.ucsd.edu) & DataHub (datahub.ucsd.edu)
- Take the incoming survey to vote on an office hour time
- (Optional) Sign up for Cogniterra and/or DataQuest
- (Super Duper Optional) Install VS Code

You only *really* need access to the DataHub, but having the ability to use Python & Jupyter Notebooks on your local computer *may* be useful (especially for final projects)!

To interact with Jupyter Notebooks on your computer using VS Code *OPTIONAL*

1. Install VS Code for your operating system.
2. If you're using Windows, [download git](#).
3. In Terminal (Mac) or the Anaconda Prompt (Windows), clone the repository by running the following command:
git clone [http://GITHUBURL](#)
4. File > Open Folder > Find the folder you just created
5. Install ipykernel when prompted